

Quiz

Apr. 20, 2026

Which of the following statements accurately describe the Buddy System as implemented in Linux memory management?

- A** The Buddy System manages free physical memory as a collection of blocks whose sizes are powers of two (e.g., 2^N pages).
- B** When allocating memory, a free block is repeatedly divided by two until the resulting block is just large enough for the request (any further split would make it too small).
- C** Two blocks are considered “buddies” if their virtual addresses differ only in a single bit.
- D** When a block is freed, the system immediately splits it into smaller blocks to prepare for future small allocations.
- E** The Buddy System recursively merges a freed block with its buddy if the buddy is also free.

提交

Which of the following statements accurately describe the Slab Allocator?

- ☐ A A slab consists of one or more physically contiguous pages.
- ☐ B A cache is made up of one or more slabs, and there is one cache for each type of kernel object.
- ☐ C The Slab Allocator wastes significant memory due to fragmentation because it always allocates fixed-size blocks.
- ☐ D Objects are created and initialized in advance, and freed objects are marked as free for immediate reuse.
- ☐ E The Slab Allocator is the only allocator ever used in all versions of the Linux kernel.

提交

Which of the following statements about memory security features are correct?

- A** Data Execution Prevention (DEP) prevents data pages from being executed by using the NX (No eXecute) bit in page table entries.
- B** DEP makes stack pages non-executable to help mitigate buffer overflow attacks.
- C** Address Space Layout Randomization (ASLR) randomizes the virtual addresses of stacks, heaps, and libraries.
- D** ASLR is designed to defeat Return-to-libc attacks and Return Oriented Programming attacks by making the addresses of libc functions and gadgets unknown to the attacker.
- E** DEP alone is sufficient to prevent all forms of buffer overflow attacks without needing ASLR.

提交

Which of the following statements accurately describe preemptive and non-preemptive scheduling?

A

A scheduling algorithm is considered preemptive if CPU scheduling decisions take place only when a process switches from running to waiting or when a process terminates.

B

In non-preemptive scheduling, once a process acquires the CPU, it keeps the CPU until it either terminates or voluntarily enters the waiting state.

C

Preemptive scheduling allows the scheduler to take the CPU away from a running process when a new process arrives with a higher priority.

D

Shortest Job First (SJF) can be implemented as either preemptive or non-preemptive.

E

Round Robin (RR) is a non-preemptive scheduling algorithm because each process runs to completion before the next process starts.

提交

多选题 2分

Consider the following set of processes with their arrival times and CPU burst times. The system uses Preemptive Shortest Job First (SJF) scheduling. Which of the following statements about the scheduling outcome are correct?

Process	Arrival Time	CPU Burst Time
P1	0	7
P2	2	4
P3	4	1
P4	5	4

提交

A

P3 is the first process to complete execution.

B

P1 runs to completion without ever being preempted.

C

P2 is preempted by P3 at time 4.

D

The average turnaround time is 8.0.